

Wireless Networking

Task – 1 : List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Solution : Wi-Fi Standards Comparison

Wi-Fi Standard	Common Name	Frequency Band	Key Speed / Difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Up to about 600 Mbps; supports both bands.
802.11ac	Wi-Fi 5	5 GHz	Much higher throughput than 802.11n; operates on 5 GHz.
802.11ax	Wi-Fi 6	2.4 GHz and 5 GHz	Higher efficiency and capacity; supports faster speeds than Wi-Fi 5.

Key Difference: The major differences are the supported frequency bands, maximum data rates, and how efficiently the standard handles multiple devices.

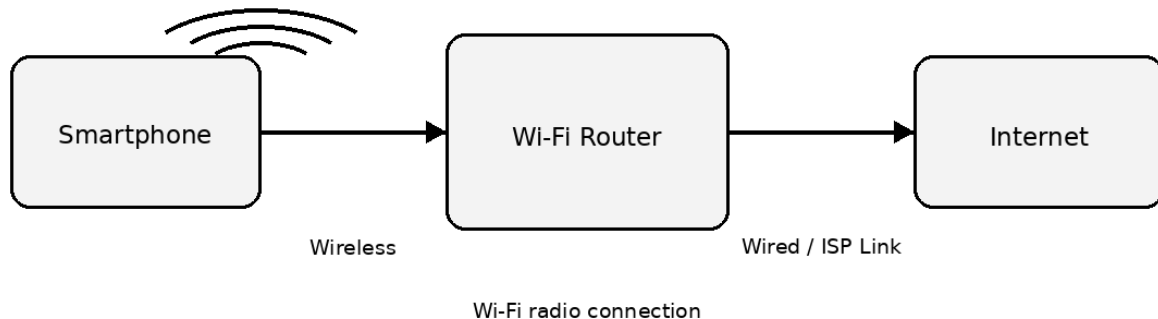
- 802.11n (Wi-Fi 4) can operate on both 2.4 GHz and 5 GHz.
- 802.11ac (Wi-Fi 5) focuses on the 5 GHz band and provides higher throughput.
- 802.11ax (Wi-Fi 6) improves efficiency in crowded networks and supports higher performance.

Task – 2 : Draw a simple diagram showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

Solution : Home Wi-Fi Connection

A smartphone connects wirelessly to the home Wi-Fi router using radio signals. The router then forwards the traffic through its wired connection to the ISP or Internet connection.

Smartphone to Internet via Home Wi-Fi Router



Wireless Part: Smartphone ↔ Wi-Fi Router. The smartphone communicates with the router over Wi-Fi.

Wired Part: Wi-Fi Router ↔ ISP/Internet. The router normally uses an Ethernet or other physical WAN link to reach the ISP equipment.

Task – 3 : Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type for at least 3 networks, and explain which one you would choose and why.

Solution : Wi-Fi Network Scan and Selection

The values below are an example format.

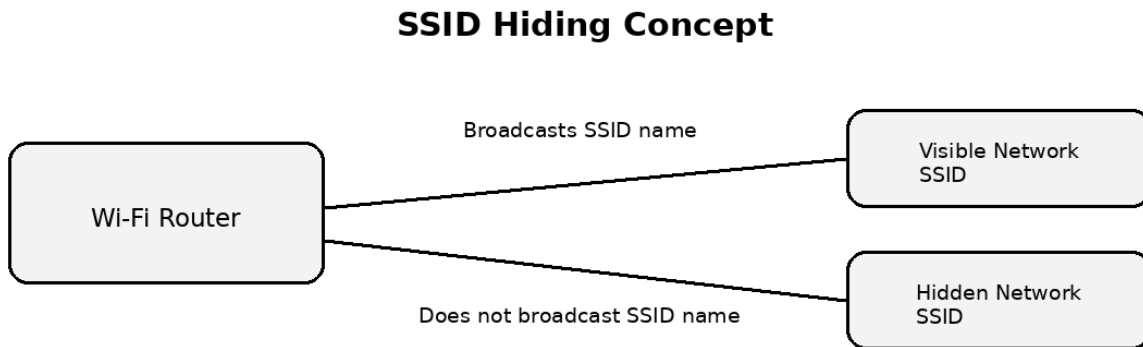
No.	SSID	Signal Strength	Security Type
1	Home_WiFi	-42 dBm (Strong)	WPA3-Personal
2	College_WiFi	-58 dBm (Good)	WPA2/WPA3
3	Guest_WiFi	-72 dBm (Weak)	WPA2-Personal

Selection: From the sample results, I would choose Home_WiFi because it combines a strong signal with WPA3-Personal security. A stronger signal generally provides a more reliable connection, while WPA3 provides modern wireless security.

Task – 4 : Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

Solution : SSID Hiding

SSID stands for Service Set Identifier, which is the name used to identify a Wi-Fi network. SSID hiding is a router setting that stops the network name from being included in normal Wi-Fi beacon broadcasts.



Advantage: Hiding the SSID can make the network less visible in the normal Wi-Fi network list. This can reduce casual discovery of the network name.

Disadvantage: SSID hiding does not provide strong security by itself. The network can still be discovered by someone using Wi-Fi analysis tools, and users may need to enter the network name manually.

A strong Wi-Fi password and modern security such as WPA2 or WPA3 are more important security controls than simply hiding the SSID.